Grading of $\mathcal{E}$:

Recall $\mathcal{E}$ has a $\mathbb{Z}^2$-grading : • any element in $\Bbbk[c_1^{\pm1}, c_2^{\pm1}]$ has degree $(0,0)$,

• $\deg(f(x^{m,n})) = (dm, dn)$ where $d = \deg f$, $f \in \Lambda$.

$\therefore \mathcal{E}^+$ is an $\mathbb{N}\times\mathbb{Z}$ graded subalgebra of $\mathcal{E}$.

Denote : $\mathcal{E}^{(a,b)}$ the (a,b)-graded component.

$$(\mathcal{E}^+)^{(\ell,0)} = \bigoplus_{d\in\mathbb{Z}} (\mathcal{E}^+)^{(\ell,d)}.$$

Grading of T:

The algebra T is $\mathbb{N}\times\mathbb{Z}$ graded, where the (ℓ,d)-component consists of Laurent polynomials $\phi \in T^\ell$ homogeneous of degree d.

(# variables; homogeneous degree)

Grading of the shuffle algebra S :

If $\Gamma(w,y)$ satisfies (*) in Theorem 3.2.1 (in particular, $\Gamma = \hat{P}$ or $\check{P}$), then σ_Γ^ℓ are homogeneous of degree 0, i.e. $\prod_{i<j}\Gamma(z_i,z_j)$ is homogeneous and with degree 0 ($\because \sigma_\Gamma^\ell(f)$ is deg-invariant for $f\in T^\ell$ homo. of deg. d)

$\therefore T\cap I$ is an $\mathbb{N}\times\mathbb{Z}$ graded ideal (same for $T\cap I_{\hat{P}}$ and $T\cap I_{\check{P}}$ b/c $I = I_{\hat{P}} = I_{\check{P}}$).

$\because$ $S = T/T\cap I$

$\therefore$ S inherits an $\mathbb{N}\times\mathbb{Z}$ grading from T

$\because$ $S_{\hat{P}} = \sigma_{\hat{P}}(T)$ and $S_{\check{P}} = \sigma_{\check{P}}(T)$

$\therefore$ $\sigma_{\hat{P}}$ and $\sigma_{\check{P}}$ induce $\mathbb{N}\times\mathbb{Z}$ gradings on $S_{\hat{P}}$ and $S_{\check{P}}$ respectively, such that

a symmetric rational function $h(z_1,\dots,z_\ell)$ in $S_{\hat{P}}$ and $S_{\check{P}}$ belongs to $S_{\hat{P}}^{(\ell,d)}$ and $S_{\check{P}}^{(\ell,d)}$ iff h is homogeneous with degree d in the variables $z_1,\dots,z_\ell$.

We thus have the following proposition:

$\psi\circ\sigma_{\hat{P}}^{-1}$ — deg invariant — $\psi\circ\sigma_{\check{P}}^{-1}$

Proposition 3.3.1: The isomorphisms $\underbrace{\psi : S \to \mathcal{E}^+}$, $\psi_{\hat{P}} : S_{\hat{P}} \longrightarrow \mathcal{E}^+$, $\psi_{\check{P}} : S_{\check{P}} \longrightarrow \mathcal{E}^+$ preserve $\mathbb{N}\times\mathbb{Z}$ grading.

- $\psi(z_1^a) = p_1[-Hx^{1,a}]$

  $\deg = (1,a)$ $\qquad$ $\deg = (1\cdot1, 1\cdot a) = (1,a)$

- $\underbrace{\psi(z_1^{a_1}\cdots z_\ell^{a_\ell})}_{\deg = (\ell,\, a_1+\cdots+a_\ell)} = \underbrace{p_1[-Hx^{1,a_1}]\, p_1[-Hx^{1,a_2}] \cdots p_1[-Hx^{1,a_\ell}]}_{\deg = (1,a_1)+(1,a_2)+\cdots+(1,a_\ell) = (\ell,\, a_1+\cdots+a_\ell)}$